IIT Jodhpur Academic Helpdesk: Final Technical Report

1. Abstract

This assignment implements a local, source-grounded academic helpdesk and compares an eight-tool single Google ADK agent against a coordinator with policy, schedule and email specialists. The system uses Ollama, ChromaDB, exact source citations, structured semester records and unsent templates. Full-v1 failures motivated bounded contract, context, applicability and handoff repairs. This report uses saved real inference artifacts; AI-assisted grading is not independent human verification.

2. Introduction and problem statement

Students need defensible academic answers, course information and professional email drafts. Similarity retrieval alone does not establish a policy, a timetable or a student's private records. The objective is correct supported assistance and explicit refusal, clarification or abstention when the task cannot be established.

3. Academic document corpus

The retained source snapshot has 24 PDFs and 111 extracted pages.
All current paths and extraction hashes match. There are 19 flagged pages.
Some official listings are intranet-only; local availability does not establish redistribution rights or latest applicability. No academic document or source manifest was changed for full-v2. See data_readiness_full_v2.json for per-source hashes.

4. RAG methodology

Existing extraction, cleaning, page-bounded character chunks, Chroma cosine retrieval and nomic-embed-text embeddings were preserved. The active development index has 215 chunks including review-flagged content. top_k remains 5 with bounded neighbour context. A query with an explicit table/section reference also searches actual indexed passages under the same programme filter. Missing references cannot be replaced by merely similar tables. Both modes use the same retrieval implementation.
Explicit references narrow excerpt selection; abbreviations and table headings/rows remain attached. Complete verified table spans retain exact source text rather than reconstructing numerical cells. These are engineering changes, not new academic facts.

5. Structured retrieval tools

CSV tools resolve identity independently of timetable availability, list recorded offerings, preserve co-instructors, count distinct offered course codes in Python and return source-backed instructor mappings. Six offerings and nine assignments are recorded; timetable rows are absent and completeness flags remain false. No official zero is inferred from missing matches. Both architectures expose equivalent tools.

6. Single-agent baseline

The baseline uses one ADK agent, current-turn tool stores, the common local model factory, shared request facts and canonical result rendering. A successful structured result is not reinterpreted as a policy chunk. Explicit unknown result IDs still fail.
For mixed tasks, a server-owned draft is preserved alongside validated policy output. This is a repaired v2 baseline; the original v1 implementation/results remain archived.

7. Multi-agent architecture

A deterministic ADK BaseAgent coordinator invokes Policy, Schedule and Email single-turn sub-agents sequentially. Policy owns retrieval/validation, Schedule owns structured lookup, and Email owns unsent drafts. The coordinator rechecks handoffs and combines canonical results in Python. Policy subtask projection prevents an email request from being mistaken for a fact that needs documentary substantiation. Shared guards continue to use the original user request.

8. Validation and refusal mechanisms

Policy answers require current-turn chunk IDs and continuous exact quotations. Source boundaries, historical sections and qualifying conditions are checked; one

bounded revision remains the limit. Header-preserving table passages are exact and complete-row verified. Structured outputs use server-owned result IDs. Missing personal facts stay placeholders; no email is sent. Nonacademic questions are refused.
Private academic records and unsupported future calendar years produce academic insufficient-evidence responses. Applicability remains unverified where sources lack current programme/cohort authority. Source fidelity is not semantic completeness.

9. Experimental methodology

The fixed 30 questions contain 10 policy, 8 schedule, 5 email, 4 out-of-scope and 3 unsupported cases, with two policy/email tasks. Nineteen answerable cases give a 57-response accuracy denominator per configuration at three repetitions. Other behaviors have separate denominators. Each request uses an isolated session/log.
Full-v2 uses temperature 0, 8192 context, maximum output 2400 and a six-model-call Runner limit. Generation models load sequentially, and each configuration is warmed consistently before measured requests. Model digests, source/index/configuration/ application/evaluator hashes and warmup timings are saved. V1's cold/warm imbalance prevents interpreting latency changes as an orchestration-only speed result.
The benchmark was also used for development; these are in-sample repair results, not estimates of generalization to unseen questions.

10. Full-v1 versus full-v2 findings

Experiment	Model	Architecture	Correct/graded	Pending	Graded accuracy
full-v1	qwen3:8b	single	15/57	0	26.32%
full-v1	qwen3:8b	multi	36/57	0	63.16%
full-v2 final	qwen3:8b	single	54/57	0	94.74%
full-v2 final	qwen3:8b	multi	57/57	0	100.0%

The first full-v2 attempt remains at evaluation/results/raw/full-v2. Remaining source-scoping and specialist-task issues were investigated before the separately saved final revision at evaluation/results/raw/full-v2/revision-2.
No old run was overwritten. Improvements combine contract handling, shared safeguards, source context, lookup parity and coordination changes. They cannot be assigned solely to multi-agent reasoning or any one fix without controlled ablations.

11. Qwen versus Llama

Qwen3:8b was executed live. Both Llama 3.1:8b configurations are explicitly skipped when unavailable. No Qwen/Llama quality conclusion is made. Install separately with ollama pull llama3.1:8b and run the identical warmup/benchmark workflow in a new namespace. Mocked unit tests are not benchmark inference.

12. Accuracy, groundedness and citation results

full-v1 / single: citation validity 85/91 (pending 0); policy-claim support 82/91; refusal 12/12; academic abstention 0/9.

full-v1 / multi: citation validity 119/128 (pending 0); policy-claim support 71/83; refusal 12/12; academic abstention 3/9.

full-v2 final / single: citation validity 144/144 (pending 0); policy-claim support 66/69; refusal 12/12; academic abstention 9/9.

full-v2 final / multi: citation validity 163/163 (pending 0); policy-claim support 85/88; refusal 12/12; academic abstention 9/9.

The evaluator distinguishes content, workflow, outer status, identity/quote fidelity, relevance and applicability. Known corpus sources outside the reviewed catalog are pending, not fabricated. Paraphrases and semantic completeness need independent review. Final review artifacts explicitly identify AI assistance and human_verified=false.
V1 citation metrics used a smaller source catalog and are not directly comparable as an isolated citation-quality improvement. No similarity/self-review score substitutes for ground truth.

13. Failure and stability analysis

V1 failures included successful tools discarded through the wrong final schema, wrong programme/table values, fragmented abbreviations, missing requested tables, omitted drafts and academic requests called nonacademic. Stored final-revision error_classification.json is automated triage, not an independently confirmed cause. Stability captures answers, minutes/percentages, citations, tool arguments, internal outputs and grading outcomes. Formatting-only variation is separated from material values/status changes. Repeated errors are not correct simply because they are stable.
Final-revision questions with variation:
single: P03
multi: P02, P03, P08

Residual final-run findings: single E05 omits the required advance-approval context in all three repetitions and is graded incorrect. Both modes repeat a flattened P09 table alongside the correctly labeled table; the evaluator rejects that duplicate claim in each repetition, although the requested17/10 values are available in the complete source table. Thus task accuracy and claim-support scores differ. This readability/validation inconsistency remains a known limitation, not a perfect-score claim. Human review should prioritize E05, P09 and policy relevance.

14. Latency and memory observations

full-v1 / single: mean 3.458 s; median 3.259 s; p95 12.288 s.
full-v1 / multi: mean 2.686 s; median 2.644 s; p95 6.021 s.
full-v2 final / single: mean 2.393 s; median 0.007 s; p95 7.508 s.
full-v2 final / multi: mean 1.775 s; median 0.009 s; p95 4.812 s.
Measured host: Linux-7.0.0-38-generic-x86_64-with-glibc2.39; RAM 30.97 GiB.
Windows/16 GB target verification: not established on this host. Hardware script observations, if present, include process-RSS estimates and separate Ollama GPU residency. RSS sums can double-count shared pages and do not certify memory capacity.
Run python -m scripts.verify_hardware --live on the actual target laptop.

15. Limitations

Independent human grading is pending. Timetable, full offering/category coverage, broader instructor coverage and latest policy applicability are unresolved. Some sources require access/redistribution review. Llama and actual Windows/16 GB tests were unavailable. Routing and qualifier checks are conservative English heuristics. Exact quotations can still be irrelevant; no universal semantic guarantee is claimed. Fixed-benchmark optimization may overfit known phrasing. Broader held-out questions and perturbations should be tested before deployment.

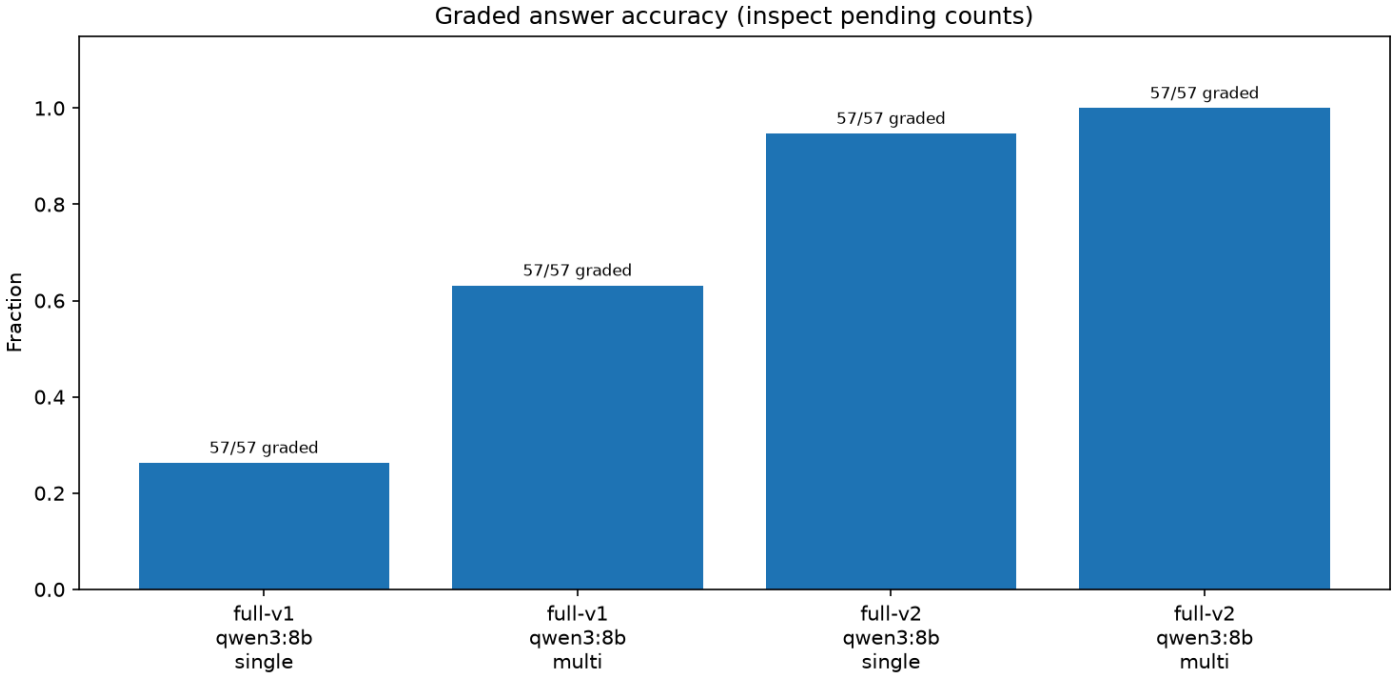
16. Conclusions

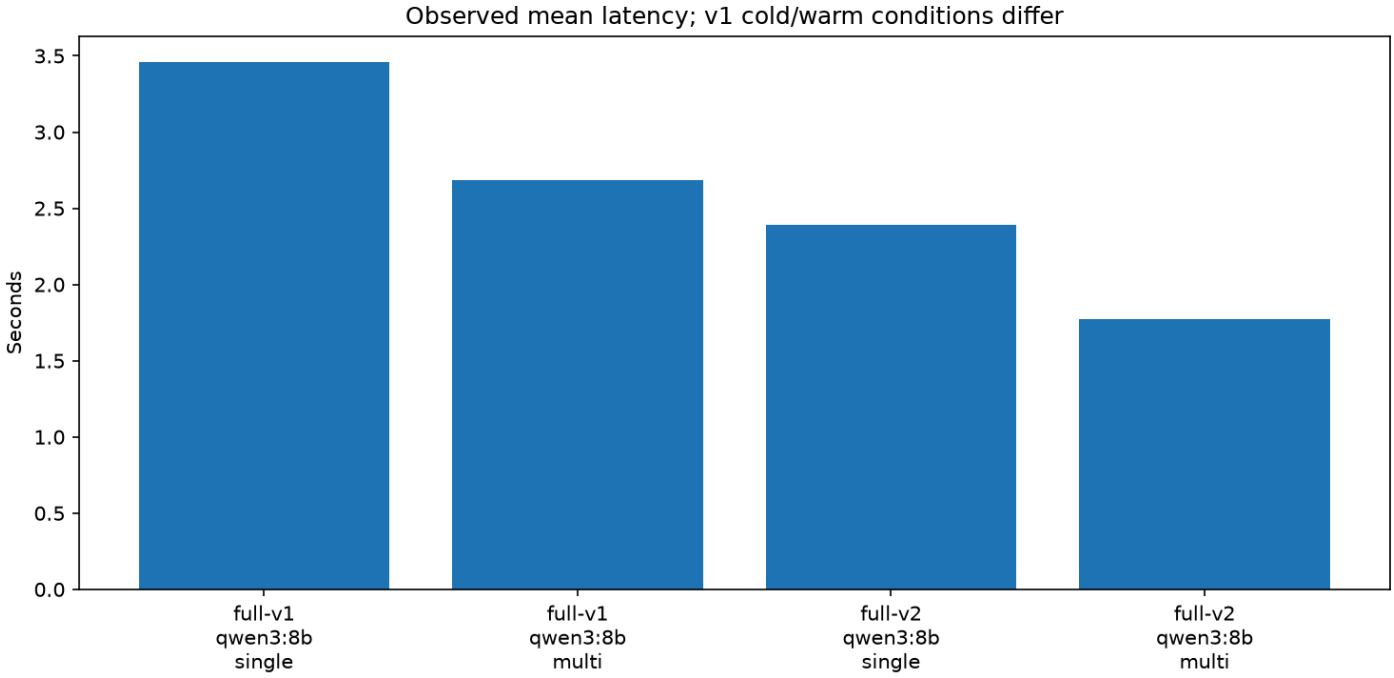
The existing architecture was retained and verified with new regressions and real sequential experiments. The recorded results quantify the effect of the implemented repair bundle on this benchmark. Missing evidence is still handled honestly.
The repository is code-complete with explicitly listed external verification gates; it is not certified submission-ready on unavailable target hardware or independently human-verified academic applicability.

17. References and reproducibility

- README.md: setup, execution, ingestion and evaluation commands.
- evaluation/questions.json and ground_truth.json: fixed benchmark and source-pinned rubric.
- reports/full_v1_preservation.json: immutable historical file hashes.
- evaluation/results/raw/full-v1/: original raw/graded benchmark.
- evaluation/results/raw/full-v2/revision-2/: final live manifests, responses and grades.
- reports/full_v2/comparison.csv and PNG charts: actual aggregate results.
- https://docs.ollama.com/windows
- https://pymupdf.readthedocs.io/en/latest/installation.html
- https://iitj.ac.in/office-of-academics/en/Academic-Regulations
- evaluation/DATA_REQUIREMENTS.md: authorized official records still required.

- Team contribution declaration: [supply actual member contributions].


Graded answer accuracy (inspect pending counts)
Fraction
0.0
0.2
0.4
0.6
0.8
1.0
57/57 graded
57/57 graded
57/57 graded
57/57 graded
full-v1
qwen3:8b
single
full-v1
qwen3:8b
multi
full-v2
qwen3:8b
single
full-v2
qwen3:8b
multi


Observed mean latency; v1 cold/warm conditions differ
Seconds
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
full-v1
qwen3:8b
single
full-v1
qwen3:8b
multi
full-v2
qwen3:8b
single
full-v2
qwen3:8b
multi